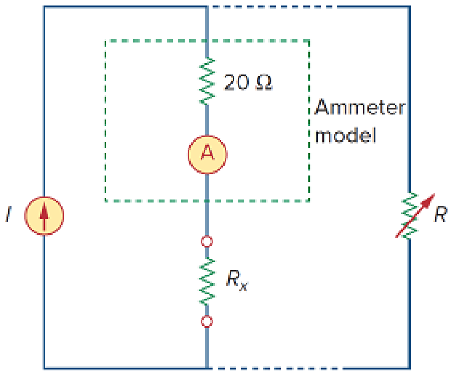
Chapter 2, Problem 73P

Problem

An ammeter model consists of an ideal ammeter in series with a 20-Ω resistor. It is connected with a current source and an unknown resistor Rx as shown in Fig. 2.133. The ammeter reading is noted. When a potentiometer R is added and adjusted until the ammeter reading drops to one half its previous reading, then R = 65 Ω. What is the value of Rx?

Figure 2.133:



Step-by-step solution

Step 1 of 2

Consider the following circuit diagram:

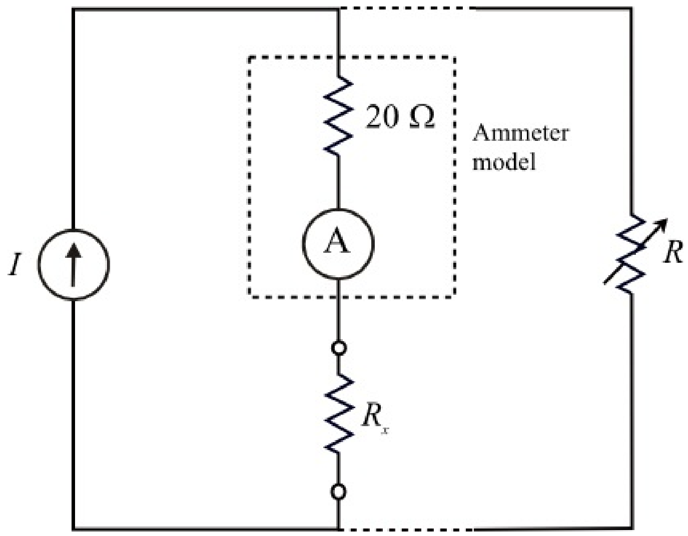


Figure 1

Step 2 of 2

Consider the following data:

Potentiometer value is, $R = 65\ \Omega$

Ammeter reading without potentiometer is I.

After adding the potentiometer, the ammeter reading is one half its previous value.

That is, $\frac{I}{2}$.

From the above statement, it is clear that the current flowing through ammeter and potentiometer is same.

Write current expressions using current divider rule.

$$I\left(\frac{20+R_x}{20+R_x+R}\right) = I\left(\frac{R}{20+R_x+R}\right)$$

$$\frac{20+R_x}{20+R_x+R} = \frac{R}{20+R_x+R}$$

$$20+R_x = R$$

$$\begin{aligned} R_x &= R-20 \\ &= 65-20 \\ &= 45\ \Omega \end{aligned}$$

Therefore, the value of R_x is $\boxed{45\ \Omega}$.